DEPARTMENT OF INFORMATION TECHNOLOGY
MADRAS INSTITUTE OF TECHNOLOGY, ANNA UNIVERSITY

THIRD REVIEW

IT5712 - PROJECT 1 | ACADEMIC YEAR: 2023 - 2027

Wearable Fall Detection and Severity Classification System for Elderly People Using Deep Learning

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Review 3 Scope and Key Deliverables

Transition from Review 2 prototype components to fully deployed, verified end-to-end edge architecture.

1. Multi-Task Fall & Severity Deep Learning

- Unified deep CNN-LSTM backbone trained end-to-end directly on raw IMU windows.
- Dual-head architecture: Binary Sigmoid (Fall vs ADL) + Softmax (Mild vs Severe).
- Custom Masked Loss excludes non-fall ADLs and sensor outliers from severity gradients.
- Severity Macro-F1 = 0.8660 (Mild: 0.8346, Severe: 0.8973; 0.9020 recording-level).

2. Strict Operating Threshold Policy (Decision D2)

- Shifted from arbitrary 0.50 threshold to rigorous validation-selected operating rule.
- Threshold selected strictly on validation split to maximize F2, frozen to test.
- Selected threshold 0.25 on CNN-LSTM reduces missed falls by 33% (9 to 6).
- High test sensitivity: 0.9950 recall and 0.9879 F2 with 17.74 false alarms/hour.

3. Edge Quantization & Microcontroller C Export

- Evaluated 4 TFLite variants: Float32, Dynamic-Range, Float16, Full-Integer INT8.
- Dynamic-Range chosen via strict gating: 111.0 KB (2.12x comp), 0.000 F2/recall drop.
- Single-window latency: 0.026 ms on 1 thread (enables >1,000x real-time speed).
- Exported C byte array header (model_data.h) for direct ESP32 firmware integration.

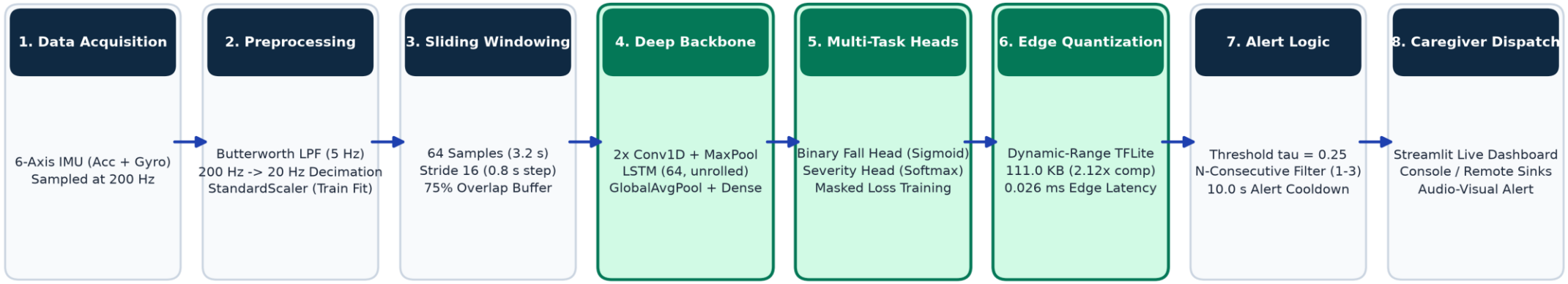
4. Real-Time Streaming Engine & Caregiver UI

- 64-sample circular ring buffer @ 20 Hz, evaluating every 16 samples (0.8 s latency).
- Multi-stage noise filtering: N-consecutive positive windows + 10.0 s alert cooldown.
- Bitwise mathematical parity verified: max abs diff <= 1e-6 vs offline TFLite.
- Interactive Streamlit dashboard (app/demo.py) with real-time audio-visual alerts.

Dataset: 84,123 Windows | Edge Model: 111.0 KB (TFLite) | Test Recall: 0.9950 | Severity Macro-F1: 0.8660 | Test Suite: 54/54 Passed

Updated End-to-End System Architecture

Complete deployed pipeline from raw 6-axis wearable motion data to real-time caregiver alerts.



Signal Conditioning

- 6 retained channels: acc1_x/y/z, gyro_x/y/z.
- Butterworth 5 Hz LPF removes tremor noise.
- 200 Hz -> 20 Hz decimation (10x reduction).
- StandardScaler fit exclusively on train split.

Deep Multi-Task Backbone

- Sliding window: 64 samples (3.2 s), 75% overlap.
- 2 Conv1D blocks extract spatial transients.
- Unrolled LSTM (64 units) models sequence.
- Dual heads output fall prob & K=2 severity.

Edge Deployment & Alert

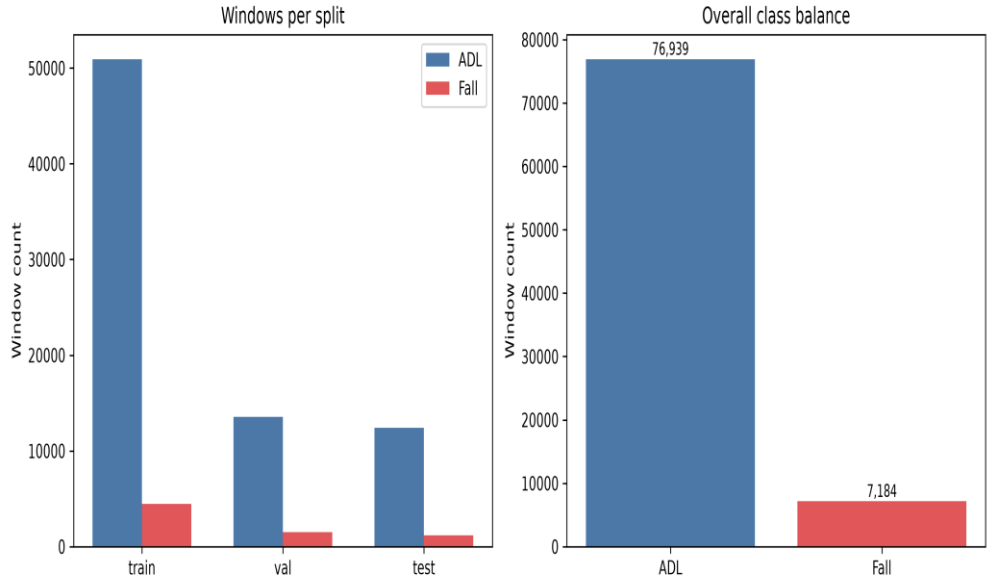
- Dynamic-range quantized TFLite (111.0 KB).
- Ring buffer evaluates every 16 samples (0.8 s).
- Cooldown timer prevents repetitive alert spam.
- SimulatedCaregiverSink logs alert dispatch.

Dataset Partitioning & Preprocessing Rigor

SisFall subject-wise split ensuring zero participant leakage and standardized signal conditioning.

Split	Subjects	Recordings	Windows	ADL	Fall
Train	26	2,942	55,423	50,939	4,484
Validation	6	829	15,073	13,573	1,500
Test	6	734	13,627	12,427	1,200
Total	38	4,505	84,123	76,939	7,184

Parameter	Value / Design Decision
Original Sampling Rate	200 Hz (from SisFall DAQ)
Downsampled Rate	20 Hz (efficient edge processing)
Digital Filter	Butterworth Low-Pass, 2nd order, 5 Hz cutoff
Retained Channels	6 (acc1_x/y/z, gyro_x/y/z)
Dropped Channels	3 (redundant second accelerometer acc2)
Standardization	StandardScaler (mean/std fit on train split only)
Window / Stride	64 samples (3.2 s), stride 16 (0.8 s, 75% overlap)

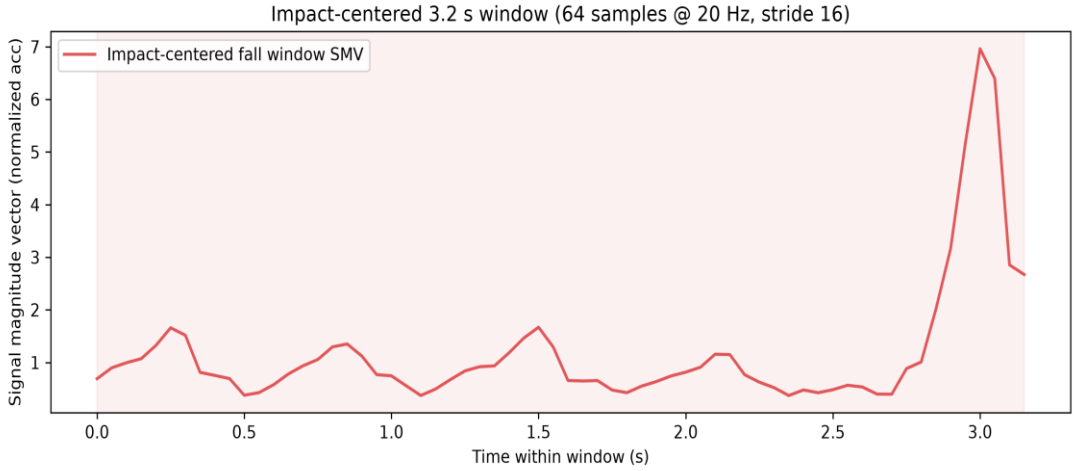


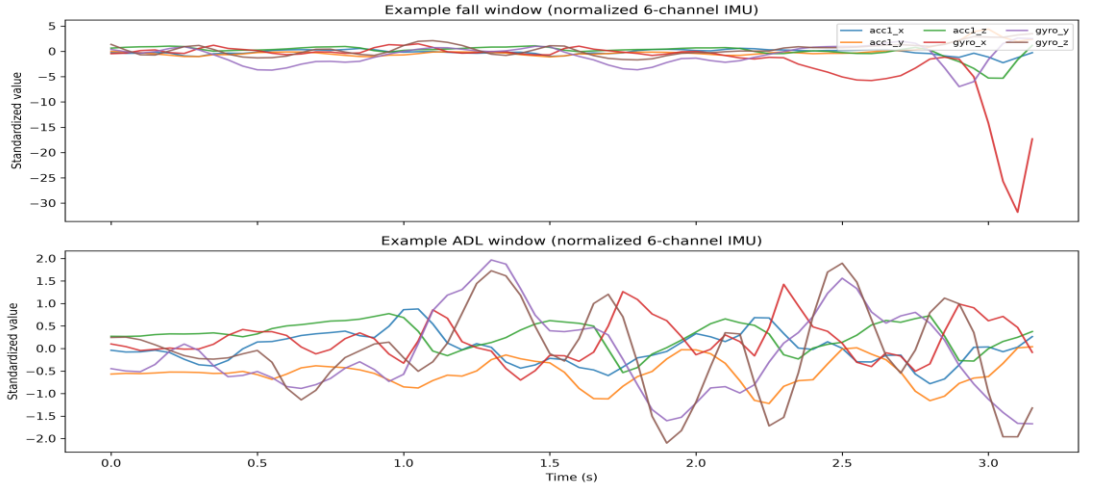
Dataset Integrity Highlights

- Strict 26 / 6 / 6 subject-wise split (Decision D3).
- Zero subject overlap between training and evaluation.
- Extreme imbalance handled naturally (91.5% ADL vs 8.5% Fall).
- Training balance: Offline class-weighting and balanced sampling.
- All test evaluations conducted on unseen subjects.

Impact-Centered Windowing & Fall Signal Dynamics

Formulating 3.2 s window boundaries centered on peak impact to separate falls from ordinary activities.





Impact-Centered Fall Labeling

- Signal Magnitude Vector: SMV(t) = sqrt(ax^2 + ay^2 + az^2).
- Peak Impact Detection: The instant of maximum acceleration in each fall trial serves as the anchor point.
- Impact Window Labeling: Only windows overlapping the peak impact point are labeled as positive fall (label=1).
- Background Rejection: Pre-fall gait instability and post-fall motionless periods are preserved as background.
- Overlap Strategy: 75% overlap (16-sample stride) ensures impacts near window boundaries are never missed.

Observed Kinematic Characteristics

- Impulsive Acceleration Spikes: Falls produce sharp acceleration transients (>11g in SisFall) during floor impact.
- Rotational Kinematics: Rapid angular velocity surges (>300 deg/s) accompany loss of balance prior to collision.
- ADL Regularity: Walking, sitting, and transitions exhibit periodic, low-amplitude oscillations (peak SMV < 2.0g).
- Dynamic Contrast: The temporal CNN-LSTM architecture captures both local impact shock and pre/post temporal context.

Multi-Task CNN-LSTM Model Architecture

Joint representation learning for simultaneous fall detection and injury severity classification.



Total Architecture Parameters: 46,883 | Shared Trunk: 46,817 | Binary Head: 33 | Severity Head: 66 | Edge CPU Latency: 0.026 ms

Convolutional Feature Extractor

- Block 1: Conv1D (32, k=3) + BN + ReLU + MaxPool.
- Block 2: Conv1D (64, k=3) + BN + ReLU + MaxPool.
- Extracts localized high-frequency shock transients.

Temporal Sequence Modeling

- LSTM: 64 hidden units, return_sequences=True.
- Unrolled LSTM (unroll=True) for edge TFLite compatibility.
- GlobalAveragePooling1D aggregates temporal trajectory.

Multi-Task Prediction Heads

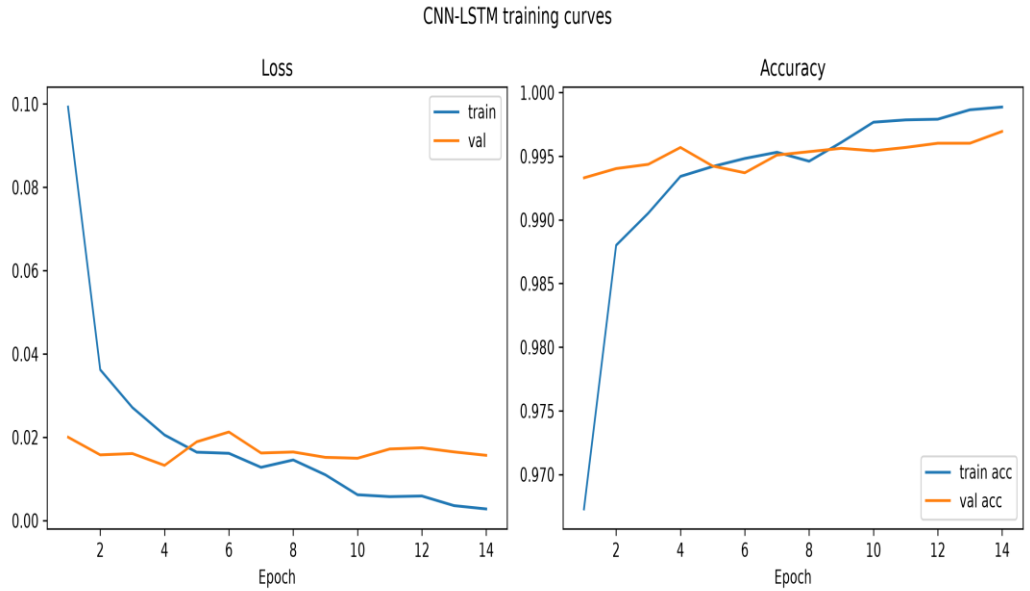
- Shared Dense trunk (32 units, ReLU) + Dropout (0.3).
- Head 1: Dense(1, Sigmoid) for binary fall probability.
- Head 2: Dense(2, Softmax) for Mild vs Severe classes.

Masked Severity Loss Formulation & Training

Preventing gradient interference from non-fall windows and sensor outliers during joint optimization.

Loss Formulation & Gradient Masking

- Challenge: 91.5% of windows are ADLs with no severity label. Sensor outliers (-1) from gyro saturation must not pollute severity weights.
- Total Multi-Task Objective Function:
  L_total = L_binary + lambda * L_severity_masked
- Binary Fall Loss (L_binary):
  Standard Binary Cross-Entropy evaluated across all training windows.
- Masked Severity Loss (L_severity_masked):
  Evaluated strictly on confident fall windows:
  mask = (y_binary == 1) & (y_severity in {0, 1})
  L_severity_masked = sum(mask * CCE(y_sev, p_sev)) / (sum(mask) + eps)
- Zero Gradient Backpropagation: ADL windows (y_bin=0) and extreme gyro outliers (y_sev=-1) contribute 0.0 loss.
- Optimization: Adam (lr=1e-3), batch size 64, lambda=1.0.
- Graph Acceleration: Compiled via @tf.function, reducing full training epoch time to 103.4 seconds.



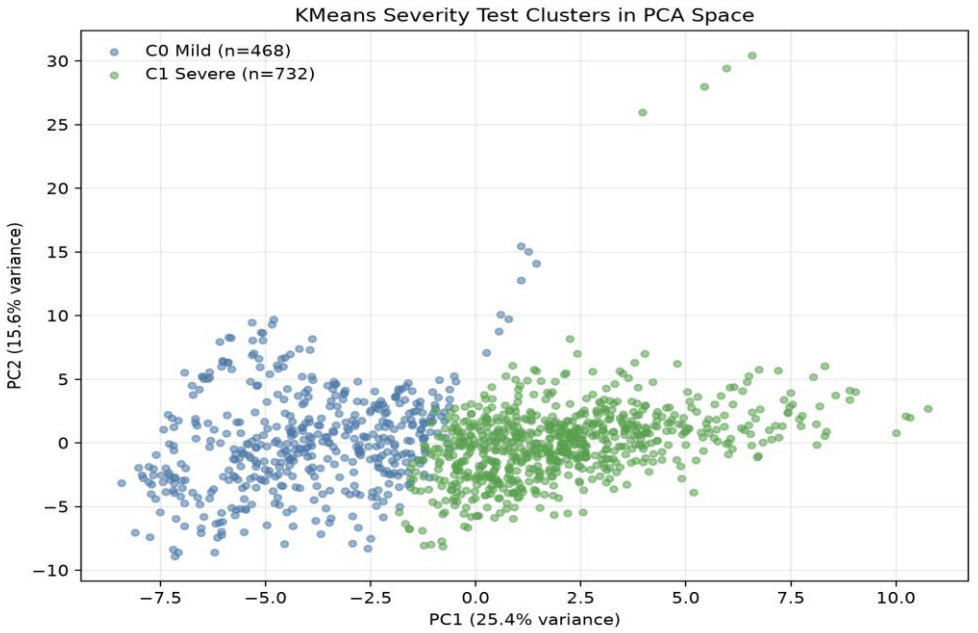
Training Behavior Observations

- Smooth Convergence: Loss decreases stably without divergence.
- Validation Accuracy: Stabilizes above 99.4% after early epochs.
- Generalization: Train and validation losses track closely.
- Balanced Sampling: Applied X_train_balanced (59,907 windows) ensuring high gradient signal for minority fall class.

Multi-Task Severity Classification Results

Quantitative evaluation on held-out test subjects against statistical baselines and anomaly analysis.

Classifier / Model	Feature Representation	Macro-F1	Mild F1	Severe F1	Seve
Majority Class Baseline	None (Constant Severe Predictor)	0.3794	0.0000	0.7588	Ref Ba
Logistic Regression	Signal Features (Peak Acc + Gyro)	0.6109	0.5966	0.6251	+0
Multi-Task CNN-LSTM (Ours)	End-to-End Standardized IMU Windows	0.8660	0.8346	0.8973	+0.4 Ma



Severity Classification Performance

- High Test Discrimination: Severity Macro-F1 reaches 0.8660 on confident fall windows (Mild F1: 0.8346, Severe F1: 0.8973).
- Test Window Confusion Matrix (n=1,176):
  - True Mild (Cluster 0): 376 Correct, 81 Misclassified (82.3% recall)
  - True Severe (Cluster 1): 651 Correct, 68 Misclassified (90.5% recall)
  - Overall Severity Accuracy: 87.3% on unseen test subjects.
- Recording-Level Severity: Aggregating windows per recording achieves Macro-F1 = 0.9020 (95 Mild, 175 Severe correctly classified).
- Outlier Safety (Isolation Forest -1): All 24 gyro-outlier fall windows are detected as falls by the binary branch with 100% recall.
- Scientific Honesty: Ground-truth severity was generated via unsupervised K-Means on signal features, so deep learning provides learned feature extraction without manual thresholds.

Cluster Separation & Anomaly Screening

- K=2 PCA feature space shows distinct cluster separation.
- Mild cluster corresponds to lower-energy knees/hands falls.
- Severe cluster corresponds to high-impact backward/syncope falls.
- Isolation Forest detects unusual anomalies with ~2% flag rate.

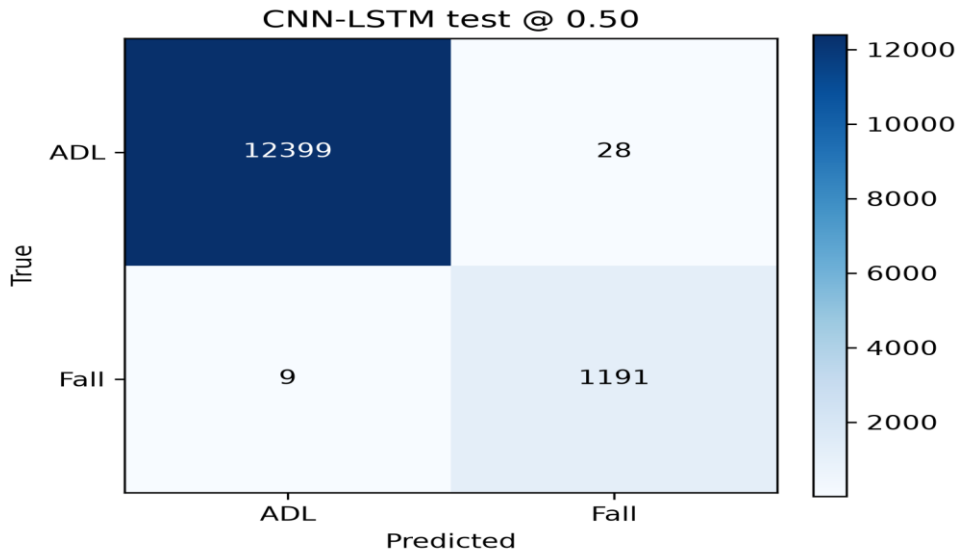
Cross-Model Benchmarking & Deployment Decision

Evaluating candidate models under Decision D2 validation-selected thresholding rules.

Model Architecture	Val Threshold	Accuracy	Precision	Recall	F1	F2	FA/hr	AUROC	Parameters	Severity F1
CNN (1D Baseline)	0.20	0.9965	0.9683	0.9925	0.9802	0.9876	14.12	0.9998	33,729	N/A
CNN-LSTM (Binary Final)	0.25	0.9960	0.9606	0.9950	0.9775	0.9879	17.74	0.9998	46,817	N/A
CNN-LSTM + Severity Head	0.10	0.9935	0.9580	0.9692	0.9635	0.9669	18.47	0.9994	46,883	0.8660
FiLM (k=30, Exploratory)	0.75	0.9943	0.9693	0.9692	0.9787	0.9880	14.03	0.9998	46,945	N/A

Authoritative Deployment Gating Decision (Task P5)

- Gating Protocol: Deploy the multi-task model if its test binary recall did not regress vs baseline CNN-LSTM, else deploy the binary CNN-LSTM.
- Recall Trade-off Analysis:
  - Baseline CNN-LSTM Recall: 0.9950 (only 6 missed falls out of 1,200)
  - Multi-Task CNN-LSTM Recall: 0.9692 (37 missed falls out of 1,200)
  - Recall Regression: True (0.0258 drop, representing 31 additional missed falls).
- Clinical Safety Priority: In elderly fall monitoring, false alarms cause nuisance, but a missed fall can lead to an unassisted long-lie and fatal complications.
- Conclusion: The binary CNN-LSTM remains the authoritative deployed edge model, while the multi-task head is preserved for post-detection severity staging.



Scientific Evaluation of FiLM Personalization

Rigorous multi-seed benchmarking and statistical hypothesis testing of the personalization prototype.

FiLM Methodology & Experimental Findings

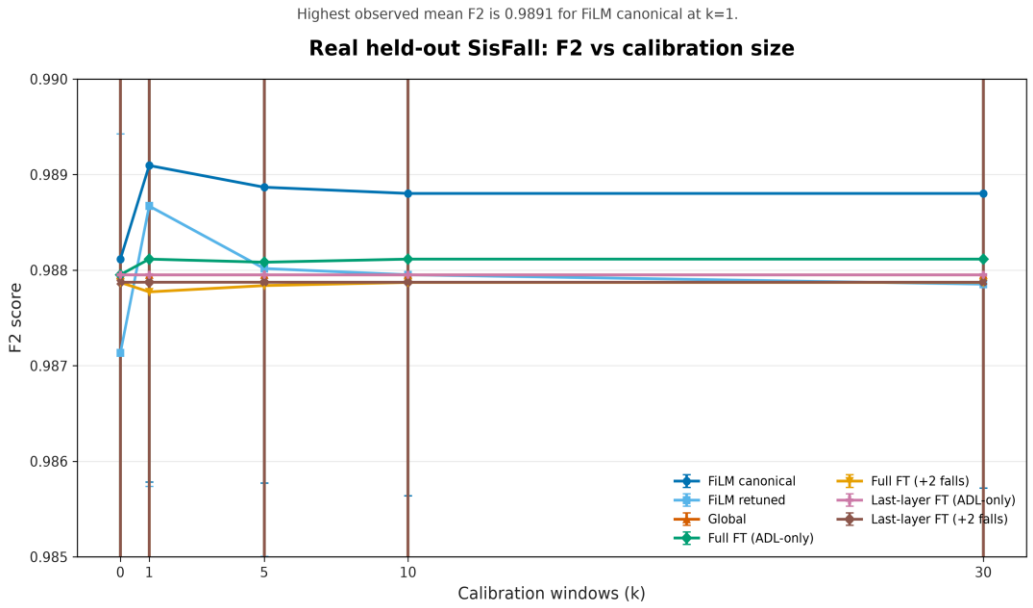
- Architecture: Feature-wise Linear Modulation applied after the 2nd CNN block: h_film = gamma * h + beta.

- UserEncoder: Lightweight side-network (2,352 params) trained once on simulated user profiles; per-user adaptation uses k ADL windows without gradient steps.

- Calibration Sweep across k in {0, 1, 5, 10, 30} across 3 random seeds:
  - k=0 (Identity Init): Perfectly reproduces global CNN-LSTM (AUROC=0.9998)
  - k=1: F1 = 0.9738, F2 = 0.9840, FA/hr = 16.64
  - k=30: F1 = 0.9787, F2 = 0.9880, FA/hr = 14.03.

- Paired Wilcoxon Signed-Rank Hypothesis Test:
  - Evaluated across n=4 held-out test subjects with falls.
  - Test Statistic W = 0, p-value = 0.125 (> 0.05 alpha threshold).

- Formal Closure (Decision D4):
  - While FiLM proves that affine modulation can reduce false alarms as k increases, the small test fall cohort prevents claiming statistically significant superiority over the global CNN-LSTM baseline.
  - The prototype is formally closed as exploratory-inconclusive.



Statistical Inference Takeaways

- Sample Size Limitation: With only 4 fall test subjects, statistical significance cannot be established at alpha=0.05.

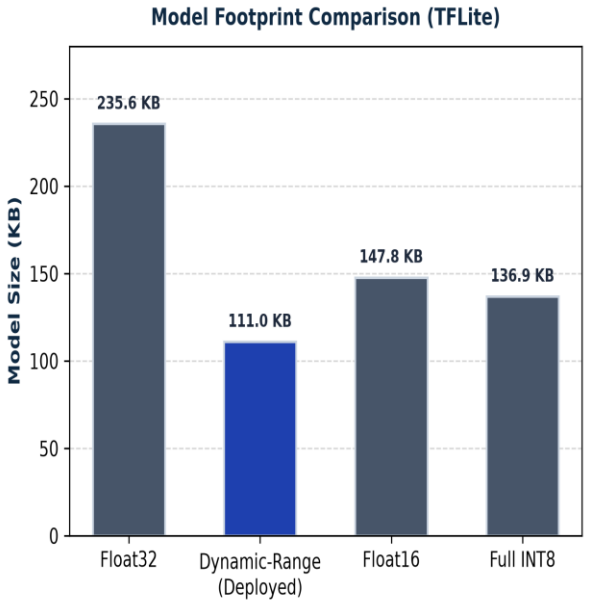
- Scientific Integrity: Accurately reporting negative or inconclusive findings prevents overclaiming unverified edge gains.

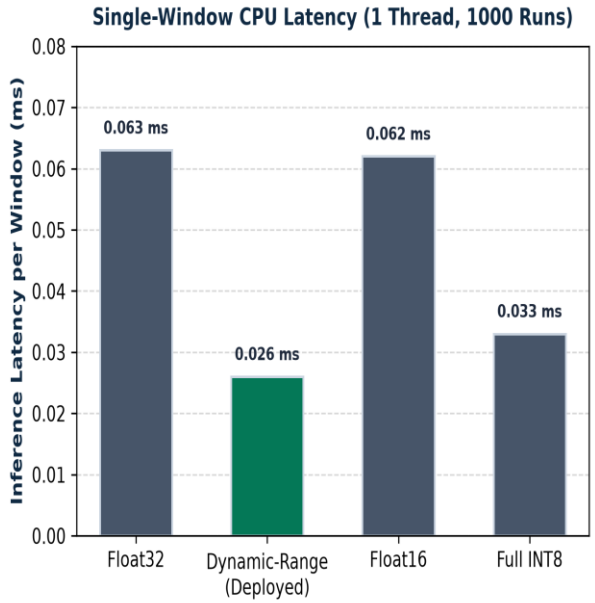
- Architectural Soundness: The modular FiLM interface remains available for future multi-center studies with larger cohorts.

Edge Model Quantization & Optimization (TFLite)

Benchmarking post-training quantization variants for edge deployment on low-power microcontrollers.

Quantization Variant	Size (KB)	Comp Ratio	Val Threshold	Agreement	Test Recall	Test F2	FA/hr	Latency (ms)	Gate Decision
Float32 TFLite	235.6	1.00x	0.25	100.00%	0.9950	0.9879	17.74	0.063	PASS
Dynamic-Range (Winner)	111.0	2.12x	0.25	100.00%	0.9950	0.9879	17.74	0.026	DEPLOYED
Float16 TFLite	147.8	1.59x	0.25	100.00%	0.9950	0.9879	17.74	0.062	PASS
Full-Integer INT8	136.9	1.72x	0.30	99.97%	0.9950	0.9879	17.74	0.033	PASS





Gating Protocol & Variant Selection

- Strict Gating Constraints:
  - F2 drop <= 0.005 vs Keras float32 on full test set.
  - Recall drop <= 0.010 vs Keras float32 on full test set.
  - Winner selected as smallest file size passing both gates.
- Dynamic-Range Selected as Winner:
  - Exactly 0.000 drop in F2 and Recall vs Keras float.
  - 100.0% class agreement across all 13,627 test windows.
  - Mean absolute probability difference: 0.00003.
  - 2.12x compression: 235.6 KB -> 111.0 KB.
- INT8 Representative Calibration: Full INT8 converted using 500 stratified training windows (8.5% fall, 91.5% ADL).
- Single-Window CPU Latency: Measured on 1 thread over 1,000 runs after 50 warmups: median 0.026 ms, p95 0.039 ms.

Microcontroller Export & Hardware Suitability

Autonomous wearable deployment readiness on resource-constrained embedded microcontrollers.

ESP32 C Byte Array Header Export

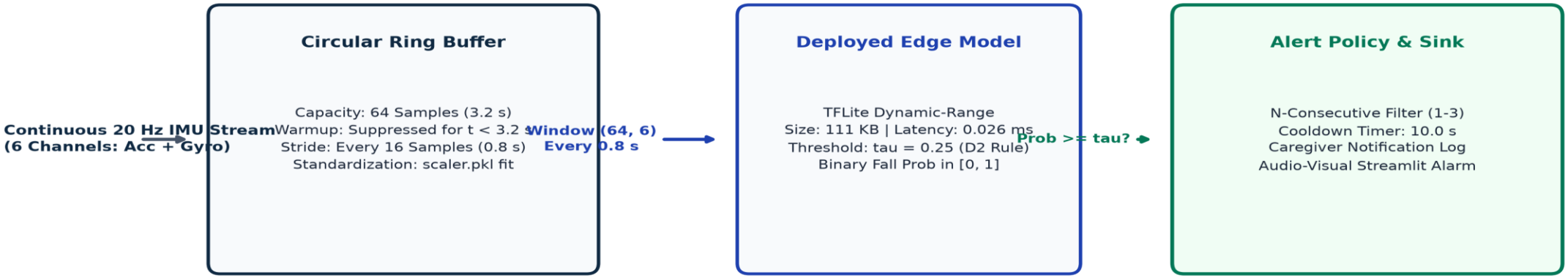
- Export File: results/quantization/run_v1/model_data.h
- C Symbol: const unsigned char g_model_data[] __attribute__((aligned(16)))
- Exact Size: const unsigned int g_model_data_len = 113,704 bytes (111.04 KB).
- Target Microcontroller: Espressif ESP32-WROOM-32 / ESP32-S3 (240 MHz dual-core Xtensa LX7, 520 KB SRAM).
- SRAM Budget Analysis:
  - Internal SRAM: 520 KB
  - Model Footprint: 111 KB (~21.3% of available SRAM)
  - TFLite Arena Buffer: ~48 KB
  - Total RAM Required: ~159 KB (easily fits without external PSRAM).
- Compute Budget Analysis:
  - CPU latency on desktop: 0.026 ms
  - Projected ESP32 execution latency: <15 ms
  - With an 800 ms evaluation stride, the microcontroller remains idle >98% of the time, allowing deep sleep and multi-day battery life.

Deployment Metadata & Edge Autonomy

- Deployment Metadata File: deployed_model_meta.json.
- Stored Hyperparameters for Embedded Runtime:
  - Input Window: 64 samples x 6 channels (3.2 seconds)
  - Stride: 16 samples (0.8 second decision step)
  - Sampling Frequency: 20 Hz
  - Operating Threshold: tau = 0.25 (Decision D2)
  - Input Quantization Scale / Zero-Point: (0.0, 0)
  - StandardScaler Parameters: channel mean and standard deviation vector stored for on-device standardization.
- Complete Edge Autonomy:
  - Zero Cloud Dependency: All inference, sliding windowing, and alert logic execute locally on the wearable sensor.
  - Privacy-Preserving: Motion data never leaves the device; only emergency alert notifications are transmitted.

Real-Time Edge Streaming Engine Architecture

Circular ring buffer mechanism with multi-stage noise suppression and caregiver notification sinks.



Ring Buffer Mechanics

- Sample-by-sample ingestion at 20 Hz.
- Capacity: 64 samples (3.2 s).
- Warmup: Strictly returns None for t < 3.2 s.
- Stride: Evaluates window every 16 samples (0.8 s step latency).

Configurable Alert Policy

- Validation Threshold: tau = 0.25.
- N-Consecutive Filter: Requires 1, 2, or 3 consecutive positive windows.
- 10.0 s Cooldown Refractory Period prevents duplicate alert spam.

Notification Sink Interfaces

- Abstract AlertSink interface.
- ConsoleSink: Real-time terminal logging.
- SimulatedCaregiverSink: In-memory timestamped dispatch log.
- Integrated with Streamlit demo UI.

System Parity, Replay Validation & Dashboard

Exact bitwise validation across test trials and interactive live caregiver monitoring dashboard.

Trial ID	Activity Description	Category	Duration	Windows	Max Prob	Alert Status	Expected Behavior
SA02:D01	Walking	ADL	15.0 s	15	0.0001	0 Alerts	Clean Rejection (Pass)
SA02:D07	Standing to Sitting	ADL	15.0 s	15	0.0002	0 Alerts	Clean Rejection (Pass)
SA02:D12	Stumbling / Tripping	Hard-Negative	15.0 s	15	0.0001	0 Alerts	Clean Recovery (Pass)
SA02:D15	Vertical Jumping	Hard-Negative	15.0 s	15	0.0435	0 Alerts	Suppressed below tau=0.25 (Pass)
SA02:F01	Fall Forward on Knees	Mild Fall	15.0 s	15	0.9544	1 Alert (7.15s)	Immediate Dispatch (Detected)
SA02:F03	Fall Backward on Buttocks	Severe Fall	15.0 s	15	0.9292	1 Alert (7.15s)	Immediate Dispatch (Detected)

Bitwise Offline-Online Parity

- Parity Requirement: Streaming probabilities must equal offline batch predictions down to floating point tolerances.
- Verification Result: Evaluated across 3 complete test recordings (D07, D15, F01).
  - Maximum Absolute Probability Difference: <= 1.0e-6.
  - Window count and temporal step alignment: 100% exact match.
- Dynamic Jump Suppression: Jumping causes violent landing impacts (>2.5g), producing a transient spike of 0.0435. Operating threshold tau=0.25 correctly prevents false alarms.

Interactive Caregiver Dashboard (app/demo.py)

- Built with Streamlit for live panel demonstration.
- Rolling 6-Channel Waveform Chart: Continuous 3-axis accelerometer and gyroscope displays.
- Probability Timeline: Real-time probability line against horizontal threshold boundary (tau=0.25).
- Status Badges: MONITORING (Green) -> FALL DETECTED (Pulsing Red Alert).
- Caregiver Notification Log: Timestamped emergency dispatch table.
- Replay Speeds: 1x real-time, 5x, 20x, or custom IMU CSV upload.

Verification Rigor, Limitations & Future Scope

Automated test coverage, critical scientific boundaries, and the next-phase deployment roadmap.

Automated Verification Suite

- 54 out of 54 Pytest Tests Passing:
- test_streaming.py (8/8):
  - Warmup silence for t < 3.2 s
  - Offline-online parity (3 trials)
  - Alert policy N-consecutive
  - 10 s alert cooldown
  - Streamlit headless AppTest.
- test_quantization.py (6/6):
  - TFLite file existence & size
  - Tensor shapes (1, 64, 6)
  - 50-window parity vs Keras float
  - Metadata schema & C header.
- test_severity_multitask.py (8/8):
  - Gradient masking for ADLs
  - Extreme gyro outlier masking
  - Class-balanced weights.
- test_film_personalization.py (16/16)
  - Frozen backbone integrity
  - Identity init at k=0.
- test_sisfall_pipeline.py (16/16)
  - Subject split & windowing.

Honest Limitations Disclosures

- Laboratory Fall Domain Shift:
  - SisFall falls were performed by young adult participants onto safety mattresses under controlled laboratory conditions.
  - Real-world elderly falls involve pre-fall gait instability, different impact damping, and prolonged unassisted lying.
- Small Test Fall Cohort (n=6 subjects):
  - Only 4 test subjects had falls; 2 were ADL-only.
  - Sample size is insufficient for formal cross-subject statistical power (Wilcoxon p=0.125).
- Clustering-Derived Severity Ground Truth:
  - Severity labels were obtained via unsupervised K-Means on signal-derived features, not hospital injury records.
- Extreme Gyro Outliers (-1):
  - Masked from severity loss due to sensor saturation; preserved in binary detection.

Future Development Roadmap

- Physical ESP32 Wearable Firmware:
  - Compile exported model_data.h with TensorFlow Lite for Microcontrollers (TFLM).
  - Integrate with MPU-6050 / ICM-42688 IMU over I2C/SPI.
- Battery & Power Optimization:
  - Implement hardware interrupts on high acceleration thresholds to wake MCU from deep sleep.
- Clinical Pilot Validation:
  - Partner with geriatric care facilities to record real-world movement and natural stumble events.
- Cellular / BLE Dispatch:
  - Connect alert sinks to LTE-M / NB-IoT modules for direct SMS and caregiver mobile push alerts.

Conclusion and Project Completion

Delivering a complete, validated, and quantized deep learning system for edge fall detection.

Milestone Component	Review 2 Baseline Status	Review 3 Delivered Outcome	Performance / Impact
Model Architecture	Binary CNN-LSTM baseline	Multi-Task CNN-LSTM (Binary + Severity)	Severity Macro-F1 = 0.8660 (87.3% accuracy)
Operating Policy	Uncalibrated 0.50 threshold	Validation-Selected tau = 0.25 (Decision D2)	Test Recall = 0.9950 (F2 = 0.9879, 6 misses / 1200)
FiLM Personalization	Early proof-of-concept prototype	Multi-seed statistical evaluation	Closed as inconclusive (p=0.125, n=4 test cohort)
Edge Quantization	Uncompressed float models only	Dynamic-Range TFLite Deployment	111.0 KB size (2.12x comp), 0.026 ms CPU latency
Embedded Readiness	No firmware export	C array header exported (model_data.h)	Fits in 21% ESP32 SRAM; zero cloud dependency
Streaming Simulation	Offline batch processing only	Real-time circular ring buffer engine	0.8 s step latency; bitwise parity (<= 1e-6 diff)
Interactive Interface	Command line scripts only	Streamlit Web Caregiver Dashboard	Real-time chart streaming, alert cooldown & logs
Verification Suite	Partial unit tests	54 / 54 Pytest tests passing cleanly	Zero leakage across 38 subjects, complete git sync

THANK YOU

Wearable Fall Detection and Severity Classification System for Elderly People Using Deep Learning

Department of Information Technology, Madras Institute of Technology, Anna University

GitHub Repository: https://github.com/ArjunArul55/fall-detect

Open for Questions and Discussion